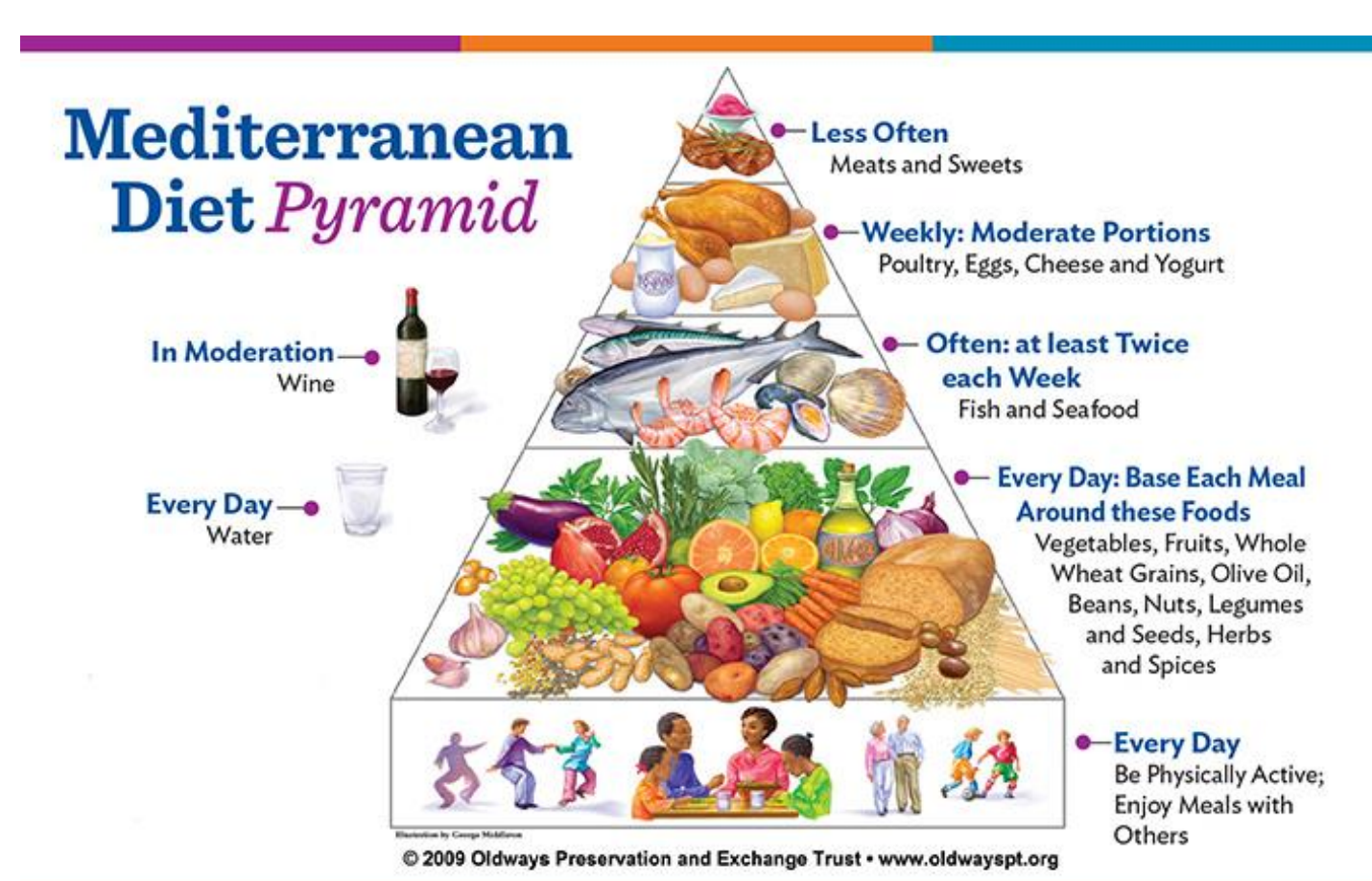


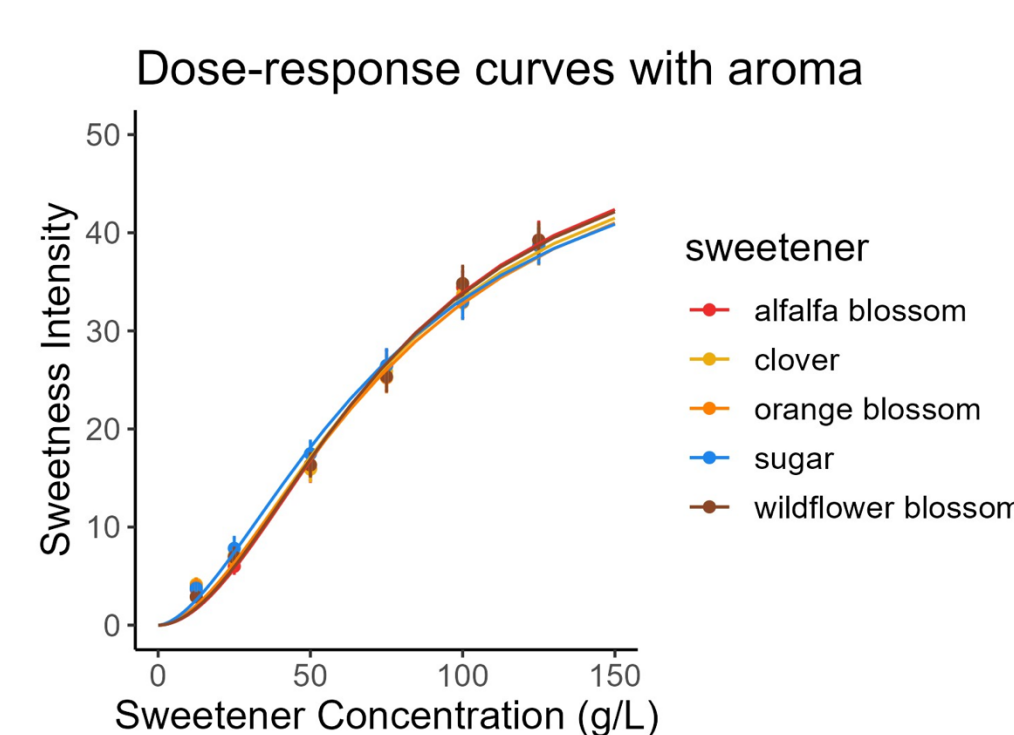
The potential of honey to increase acceptance of a Mediterranean diet

Hannah Mulheron, Aubrey DuBois, and Emily Mayhew.
Michigan State University, East Lansing, MI, USA.

Background



Regular consumption of foods from the Mediterranean diet is recommended by experts¹⁻³. However, many key foods from this diet are bitter which hinders consumer acceptance.



Sugar suppresses bitterness; however, high consumption of added sugar is a health concern^{4,5}. Previously, we found that sugar and honey are equivalently sweet by mass, providing equal sweetness with fewer calories (sugar: 3.85Kcal/g, honey: 3.04Kcal/g).

Materials & Methods

To compared the potency of honey and sugar as sweeteners when added to bitter foods, consumers rated overall liking, sweetness intensity, and bitterness intensity of foods from the Mediterranean diet that were dosed with 3 sweetener conditions.

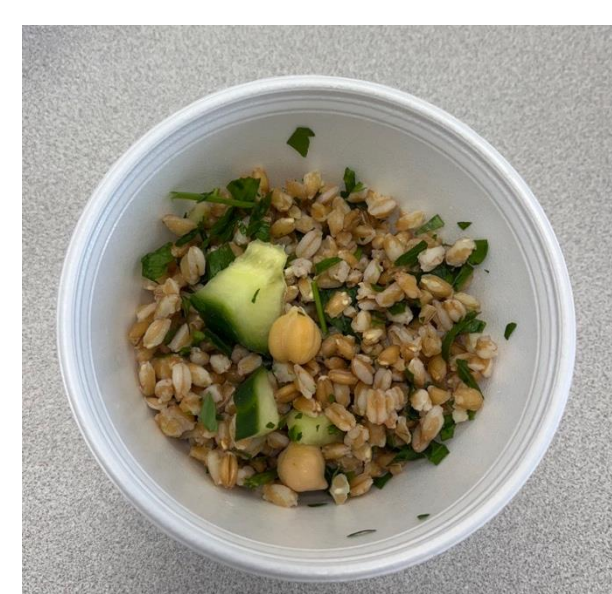
Product	Serving size	Unsweetened	Sugar		Honey	
		grams	grams	kcal	grams	kcal
Greek yogurt	¼ cup	0	4.00	15.4	4.00	12.16
Salad Dressing	2 Tbsp	0	3.15	12.13	3.15	9.58



Leafy Greens Salad
n = 101



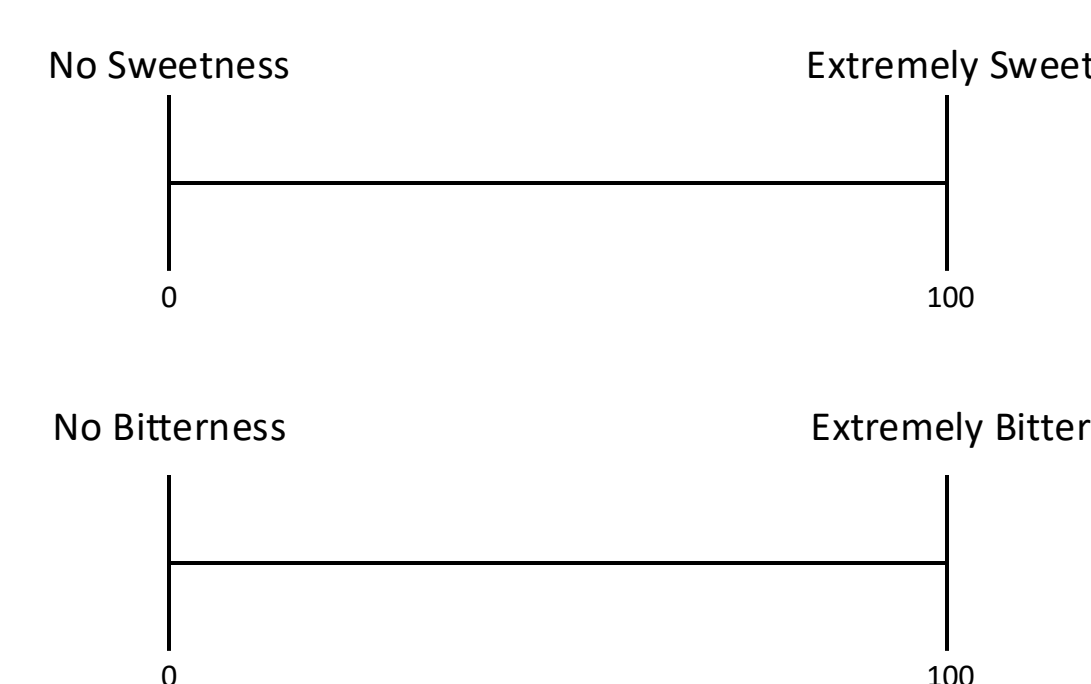
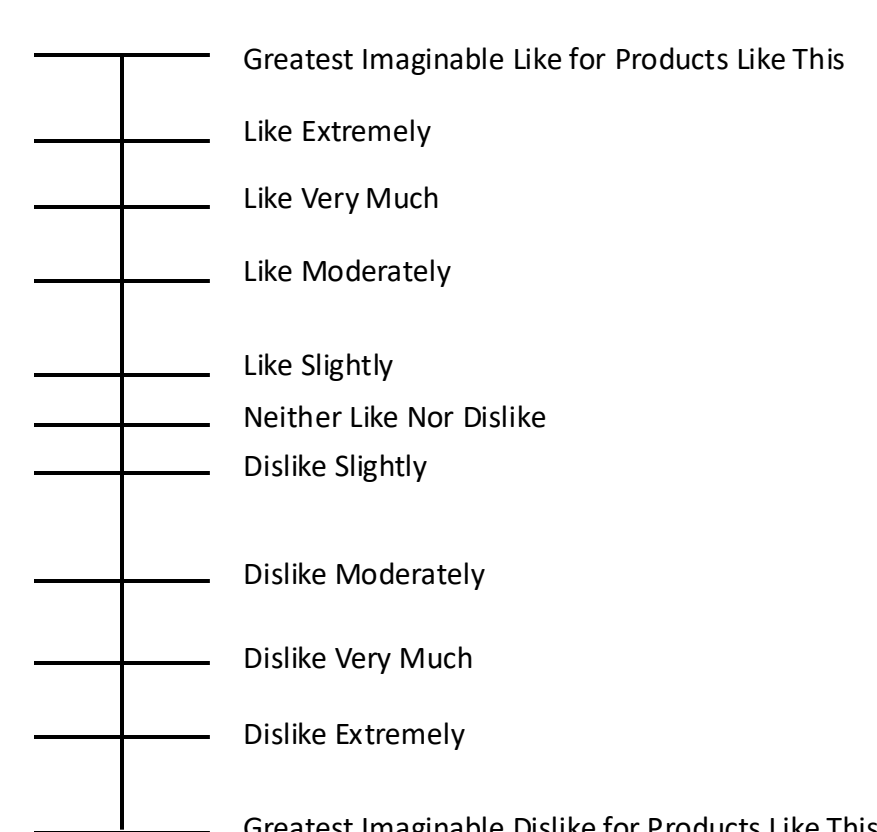
Salad dressing
n = 205



Farro Salad
n = 104



Greek yogurt with walnuts
n = 103



Results

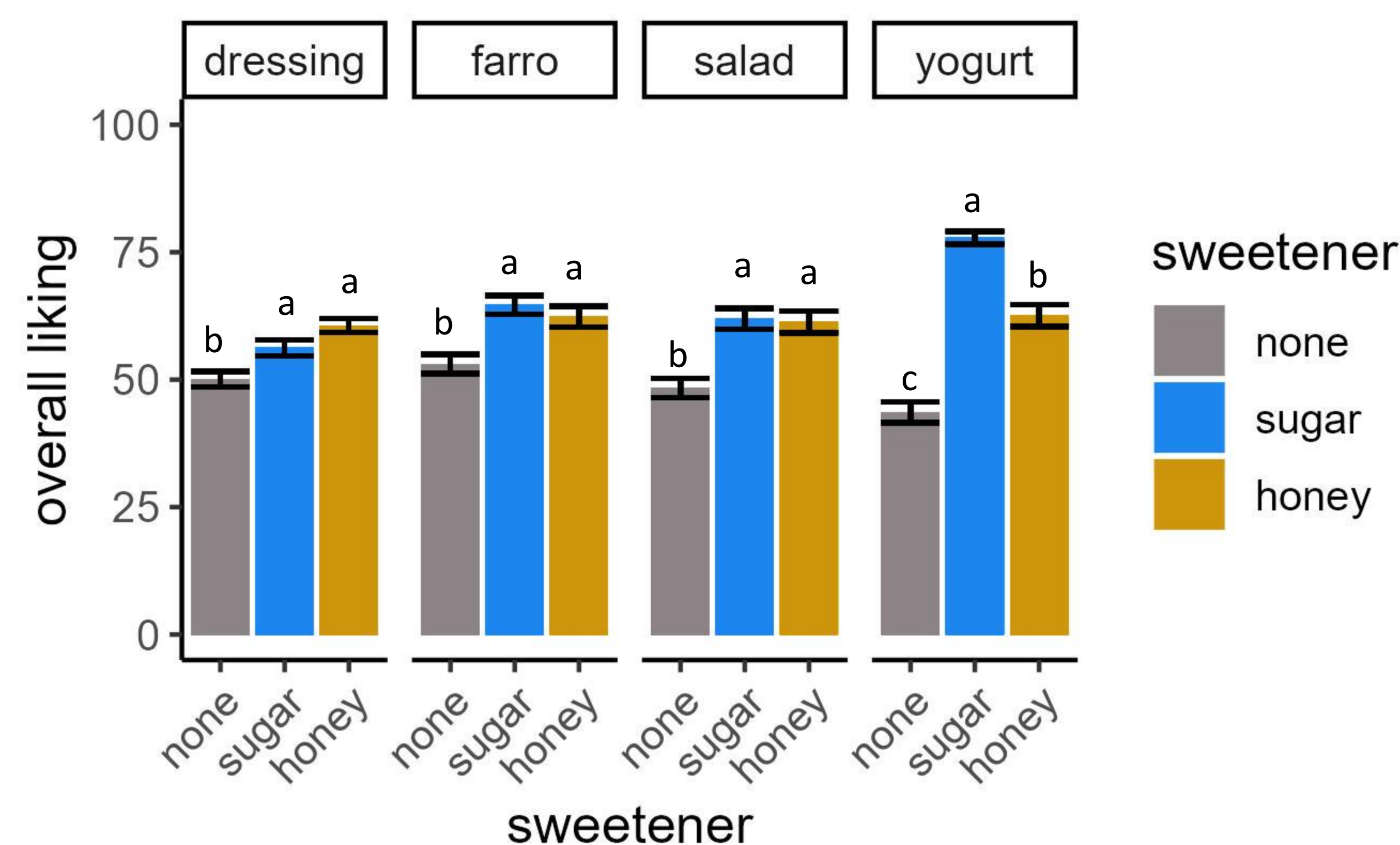


Figure 1: Mean overall liking ratings for each food tested. Fisher's LSD method used to determine post hoc means separation groups (withing each food product). Error bars represent the standard error of the mean (± 1 se).

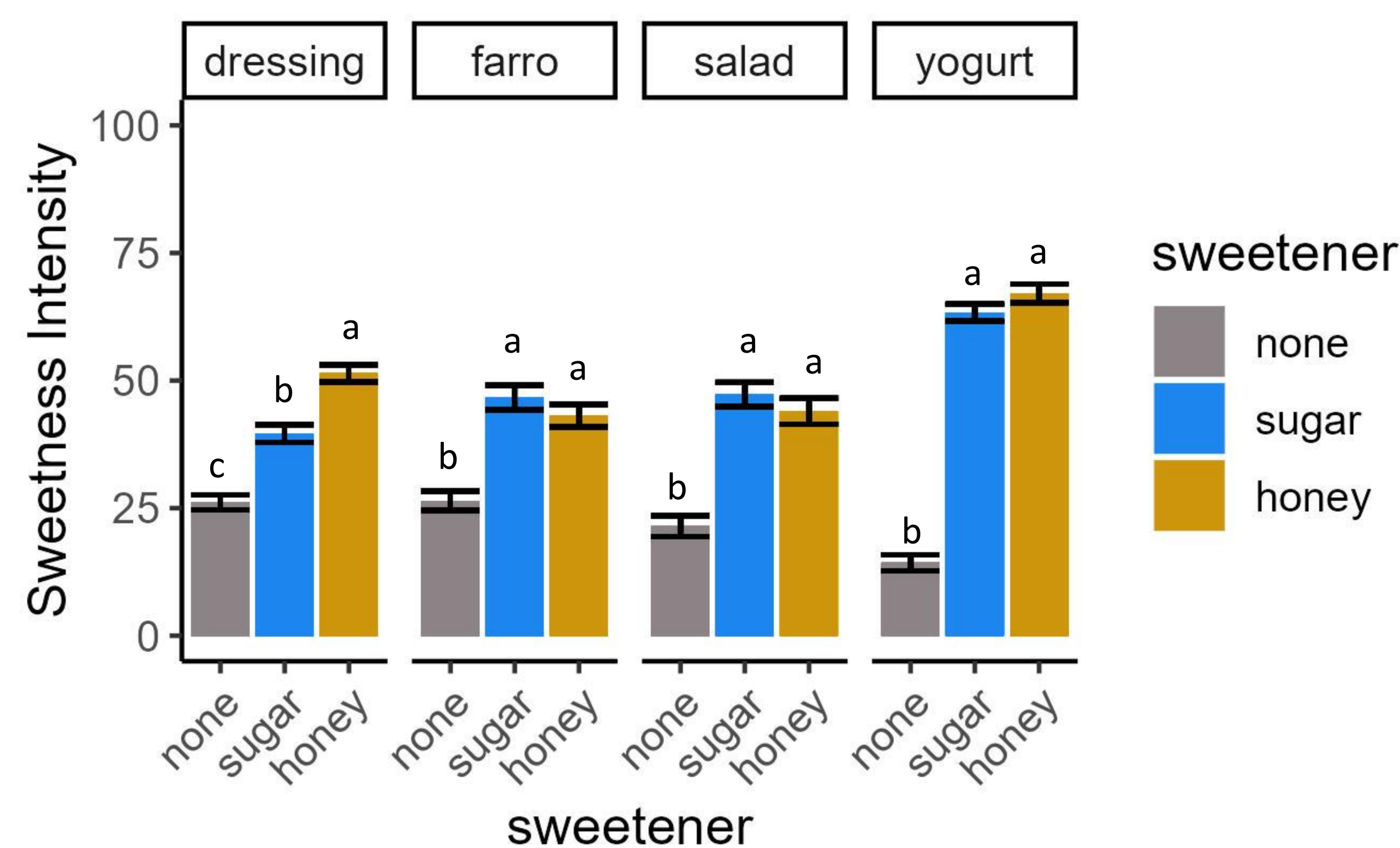


Figure 2: Mean sweetness intensity ratings for each food tested. Fisher's LSD method used to determine post hoc means separation groups (withing each food product). Error bars represent the standard error of the mean (± 1 se).

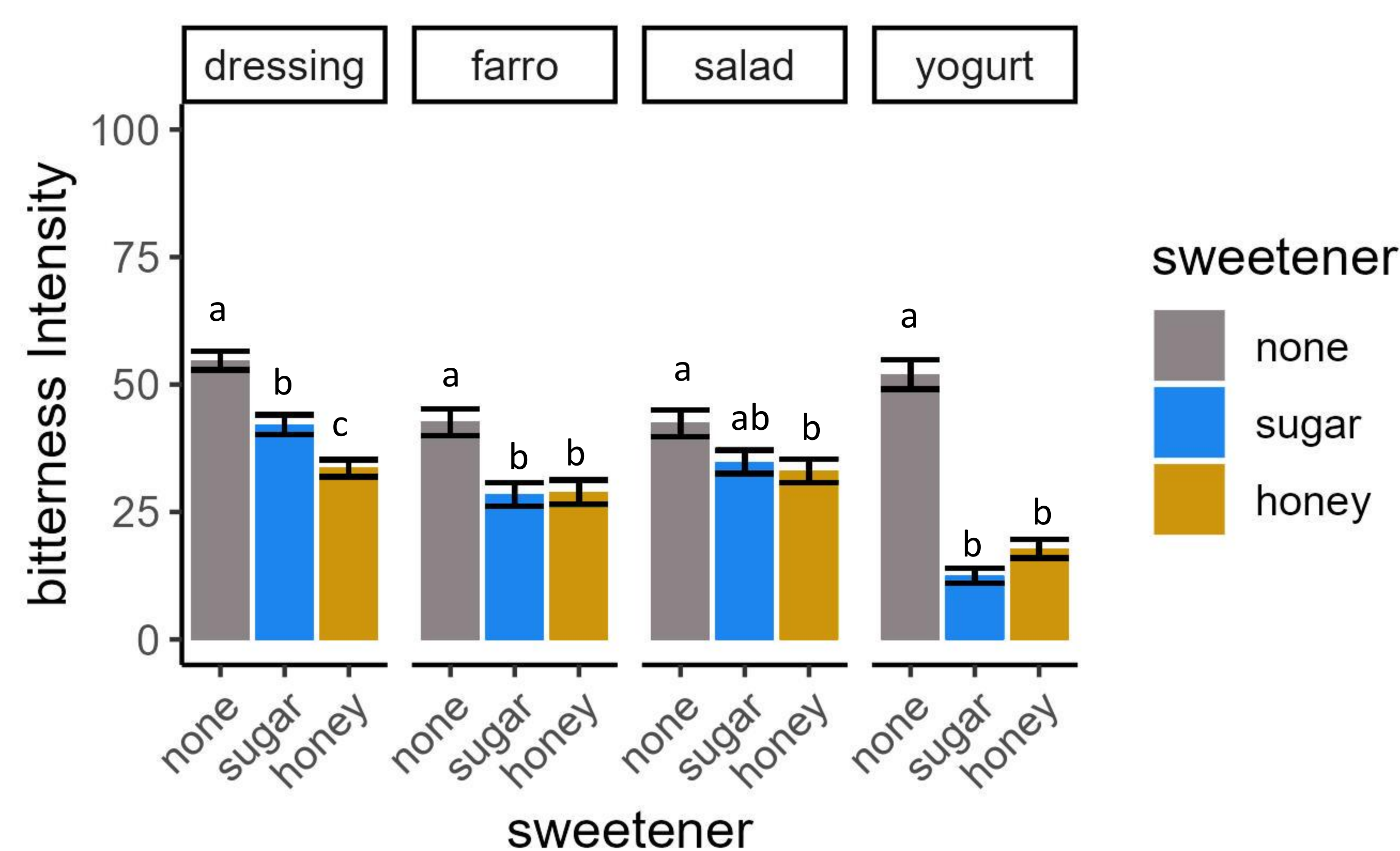


Figure 3: Mean bitterness intensity ratings for each food tested. Fisher's LSD method used to determine post hoc means separation groups (withing each food product). Error bars represent the standard error of the mean (± 1 se).

Discussion

The sweetened conditions were less bitter (except the dressed salad sweetened with sugar), sweeter, and more liked than the unsweetened condition.

Overall liking of honey and sugar conditions were liked equally for 3 of the 4 foods, sugar- sweetened yogurt was more liked than the honey condition.

Honey sweetened foods were **equally as sweet or sweeter** than the sugar sweetened conditions ; honey-sweetened vinaigrette was more sweet than the sugar condition.

Honey is on par or better than sugar at **suppressing perception of bitterness**; honey-sweetened vinaigrette was less bitter.

We hypothesis that the potency of honey as a sweetness enhancer and bitter suppressor is highly dependent on the temperature of the food that it is prepared with.

Conclusion

The sweetness enhancing potential of honey is variable depending on the composition and temperature of the food that it is added to. However, honey is on par or better than sugar at suppressing perception of bitterness. These results suggest that honey can increase acceptance of a Mediterranean diet with fewer calories from added sugar.

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